CV22FS System Hardware - Preliminary

SUMMARY DESCRIPTION

This document provides system hardware details for the Ambarella CV22FS system on chip (SoC) processor.

Guidelines for power management, the implementation of various peripheral interfaces, and PCB layout are presented.

Because best practices for camera design and board layout are strongly application-dependent, this document should be considered as a general reference guide only.

Ambarella offers board design review services as well as additional documentation to qualified customers.

KEY TOPICS

- Power on and power-off sequence information
- Pin detail for the VIN module, including sensor connection, power and I/O, and interface disconnect information
- VOUT interface detail
- GPIO pin characteristics, including I/O and pull high / low information
- Detail regarding the JTAG pins and UART interfaces
- PCB layout guidelines for DRAM, VIN, VOUT, SMIO, USB, the CV22FS power rails, and more

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1. INTRODUCTION

1.1 Introduction: Overview

This is the preliminary system hardware document for the Ambarella CV22FS system on chip (SoC) processor. Please note that the information described in this document is subject to change.

The document is intended to support CV22FS-based camera design efforts by offering guidelines for power management, the implementation of various peripheral interfaces, and PCB layout.

Note that this reference material is supplemental in nature. The document does not discuss all CV22FS interfaces or implementation cases. Because best practices for camera design and board layout are strongly application-dependent, this document should be considered as a reference guide only.

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2. RTC

2.1 Overview

This chapter provides information regarding the real time clock (RTC), and is organized as follows:

- (Section 2.2) RTC Electrical Characteristics
- (Section 2.3) RTC Disconnect
- (Section 2.4) Power-On / Off Sequence
- (Section 2.5) External Clock Source for Core and Real-Time Clock
- (Section 2.6) Watchdog Reset

Note that this document uses approximate / representative voltages to describe the CV22FS power rails in a relative sense.

Refer to the CV22FS datasheet for absolute operation range and DC characteristics.

2.2 RTC Electrical Characteristics

For additional electrical characteristics of the RTC module, please refer to the CV22FS datasheet.

2.3 RTC Disconnect

The RTC is typically maintained with one dedicated always-on power supply pin, so the clock remains active even when the core powers off. Because the RTC is an integral part of the on-chip safety mechanism, it cannot be disconnected.

2.4 Power-On / Off Sequence

This section can be used as a guide for the external power control module (PCM).

This section is organized as follows:

- (Section 2.4.1) Power-On / Off: Power-On Sequence Pin Groups
- (Section 2.4.2) Power-On Sequence Timing
- (Section 2.4.3) Power-Off-Sequence Timing

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2.4.1 Power-On / Off: Power-On Sequence Pin Groups

The CV22FS power-on pins can be divided into three groups (1, 2 and 3), which should be powered up sequentially. These power-on sequence groups are shown in the table below.

Power-On Sequence Group	Pin Name
1	**VDDI**
	AVDD
2	**AVDD18_1**
	AVDD18_2
	DDR_DVDD_IO
	DDR_DVDD_VDDQ
	OTP_VPP
	MIPI_AVDD18_IO
	VDDP
3	**AVDD33**
	NAND_VDDO
	VDDO
	SD_VDDO
	SDIO_VDDO

Table 2-1. Power-On-Sequence Groups for CV22FS Camera Applications.

Note:

DDR_DVDD_VDDQ and DDR_DVDD_IO rails as well as the 1.8-V rail for all 1.8-V domains can be turned ON/OFF simultaneously during power ON and OFF sequences.

2.4.2 Power-On Sequence Timing

This section includes general timing for power-on sequence groups, a power-on sequence timing example showing voltage and functional groups, a timing diagram illustrating the necessary alignment of the **POR_L** rising edge with the 24-MHz clock input, and the recommended circuit diagram.

The following figure illustrates the general timing for the CV22FS power-on sequence groups as defined in Table 2-1.

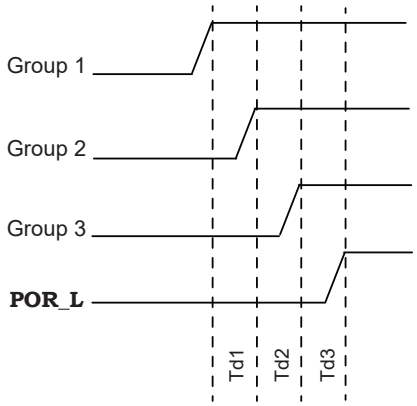


Figure 2-1. Power-On Sequence Timing for Power Rail Groups.

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Notes:

- The global reset pin **POR_L** should be asserted (low), then released or deasserted (high) after all power rails come up (Figure 2-1).
- Timing per group (**Td1**, **Td2**, **Td3** and **POR_L**) should be maintained at 10 ms or more.
- If the power ramp-up time exceeds 10 ms, the value of **Td** should be increased.

The following figure illustrates a specific example showing the power-on sequence groups (Table 2-1) and sequence timing (Figure 2-1) for representative voltage levels and functional group names.

Note that the voltage levels shown in the diagram are relative and are provided for illustrative purposes only. Please refer to the CV22FS chip datasheet for the electrical characteristics of individual Ambarella chips.

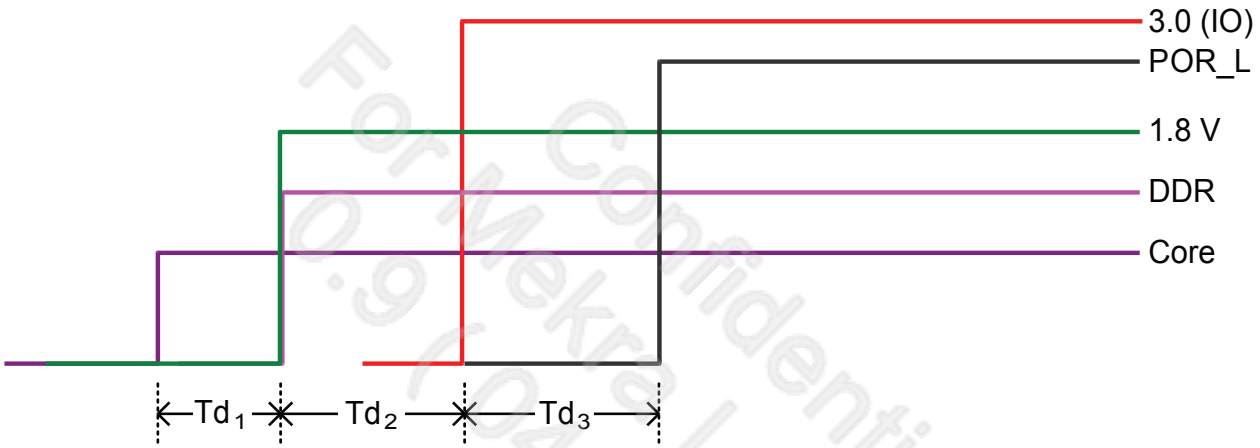


Figure 2-2. Power-On Sequence Timing Example with Representative Voltage and Functional Groups. (Refer to the Datasheet for Voltage Levels for Individual CV22FS SoCs.)

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In the above example, the power supplies are powered-up in the following sequence:

1. Core VDDI and associated 0.75 V power rails
2. DDR power rails and 1.8 V power rails
3. 3.0 V rails (IO)
4. **POR_L** external reset pin

To ensure correct boot up, **POR_L** rising edge must align with the 24-MHz clock input (see the following timing diagram).

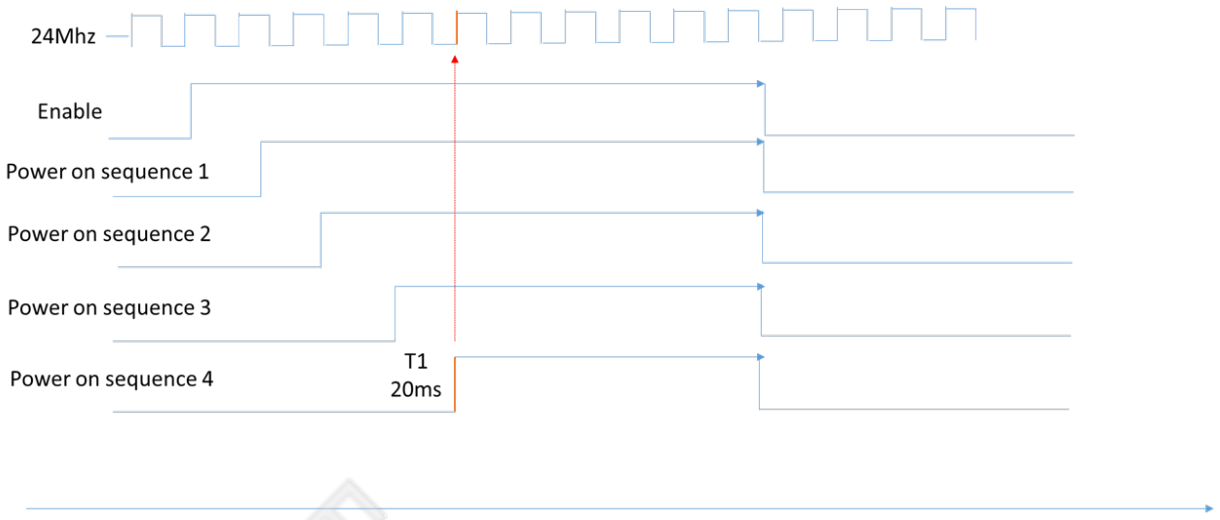


Figure 2-3. Alignment of POR_L Rising Edge with 24-MHz Clock Input to Ensure Correct Boot Up.

The recommended circuit is as follows:

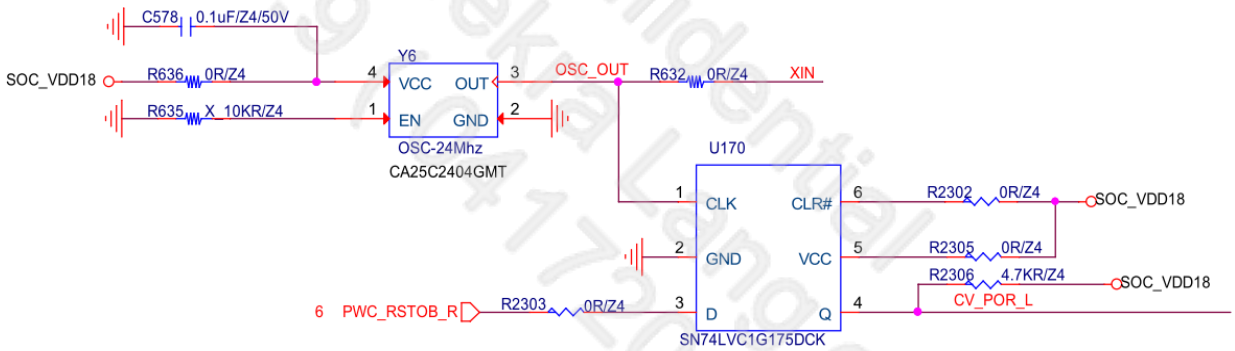


Figure 2-4. Recommended Circuit for Boot Up.

2.4.3 Power-Off-Sequence Timing

The power rails can be powered off at the same time; however, note that 1.8 V can only be powered off at the same time or later than 3.3 V, per design specification.

2.5 External Clock Source for Core and Real-Time Clock

- For the core clock that is configured to 24 MHz via an external clock oscillator, connect the 24-MHz clock with 0 V ~ 1.8 V amplitude to the **XIN** pin. Keep the **XOUT** net length as short as possible. No other loading is allowed on **XOUT**.
- For the real-time clock (RTC) with 32.768 KHz frequency driven by an external clock oscillator, connect the 32.768-KHz clock with 0V ~ 1.8V amplitude to **XI_RTC** and let **XO_RTC** float.

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- For the real-time clock (RTC) with 32.768 KHz frequency driven by a crystal oscillator, connect the 32.768-KHz crystal to **XO_RTC** and **XI_RTC** pins.

2.6 Watchdog Reset

If programmed to do so, the following pins can serve as external reset triggers for the watchdog timer (when the watchdog timer expires, the pins will assert):

- **TIMER0**
- **SSIOEN1**
- **PWM2**
- **ENT_2ND_REF**

(Search for **WDT_RST_L** in the multiplexed functions of Table 3-1 in the chip's *datasheet*.) .

Ambarella suggests connecting one of the above pins to an external MCU or CPLD. When the pin asserts, the MCU or CPLD can issue a cold reset (**POR_L**).

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3. SYSTEM BOOT, CLOCKING AND CONFIGURATION

3.1 Boot: Overview

This chapter includes the following sections:

- (Section 3.1) Boot: Overview
- (Section 3.2) Boot: Power On Configuration

3.2 Boot: Power On Configuration

The CV22FS chip uses a 32-bit power on configuration with strapping functions on external pins. After deassertion of the **POR_L** external reset pin, the chip generally latches the initial hardware settings on the pins.

The table in this section lists the power on configuration bits POC[0:31] and corresponding strapping pins. However, if POC[31] is set to 1, some (or all) of the configuration data may be obtained from OTP ROM (Section 3.2.9).

Notes:

- Unused bit—Although external pin is assigned to the bit, it is not used for a power-on configuration function. these pins should be connected to ground using the pull-down resistor.
- Reserved bit—No external pin is assigned to the bit, and it is not used for a power-on configuration function.

Additional configuration information is listed in the following subsections:

- (Section 3.2.1) POC: Pin Assignments
- (Section 3.2.2) POC: Clock Configuration
- (Section 3.2.3) POC: Boot Mode Summary
- (Section 3.2.4) POC: Force USB Boot
- (Section 3.2.5) POC: SPI NOR Boot
- (Section 3.2.6) POC: SPI NAND Boot
- (Section 3.2.7) POC: MMC Boot
- (Section 3.2.8) POC: Boot Bypass
- (Section 3.2.9) POC: OTP ROM

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3.2.1 POC: Pin Assignments

Ball	Bits	Pin	Function	Setting
G17	0	**ENET_TXD_0**	Unused	-
H18, F17, E17	3:1	**ENET_TXD_[3:1]**	IDSP / Core / DDR / Vision clock configuration	Refer to Section 3.2.2 below
E14,D14	5:4	**ENET1_TXD_[1:0]**	Boot Mode[1:0] Select	Select boot mode (SPI NOR is the CV22FS default) 2'b00 SPI NOR 2'b01 SPI NAND 2'b10 eMMC
E16	6	**ENET1_TXD_2**	Secure boot	0 Regular boot 1 Secure boot
-	7	-	Reserved	-
W20	8	**CLK_AU**	Boot bypass	0 Disable boot bypass 1 Enable boot bypass
-	9	-	Reserved	-
L17	10	**ENET_PTP_PPS**	Force USB boot	Setting this pin to high overrides the current boot mode setting and enables USB boot. This bit is always derived from the pin.
-	12:11	-	Reserved	-
Y16	13	**UART0_AHB_TX**	Boot option 0 select	Refer to Table 3-4
V12	14	**UART1_AHB_TX**	Boot option 1 select	
U14	15	**UART_APB_TX**	Boot option 2 select	
W18	16	**I2S_SO_0**	Boot option 3 select	
R19	17	**GPIO25**	Boot option 4 select	
V18	18	**I2S_WS**	Boot option 5 select	
V15	19	**SVSYNC**	Boot option 6 select	
W15	20	**SHSYNC**	Boot option 7 select	
A13	21	**ENET_GTX_CLK**	Unused	-
-	22	-	Reserved	-
-	23	-	CLOCK_SRC_MODE0	Refer to CLOCK_SRC_MODE1 This bit can only be set from OTP
-	24	-	CLOCK_SRC_MODE1	CLOCK_SRC_MODE1 Select clock sources for core and DDR clocks 00 Normal sources 01 Reference clock 10 Reserved 11 Reserved This bit can only be set from OTP
-	25	-	Reserved	
-	26	-	PERIPH_CLOCK_MODE	Controls clock to USBHOST, SD, FLASH controller. Set this bit to 1 when GCLK_CORE is < 466 MHz 0 Peripherals use GCLK_CORE/2 1 Peripherals use GCLK_CORE This bit can only be set from OTP
-	29:27	-	Reserved	-
W16	30	**PWM0**	Unused	-

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Ball	Bits	Pin	Function	Setting
Y19	31	**PWM1**	System configuration data source	Source of sys-config data 0 Config data is set by pins 1 Config data is read from on-chip OTP (except Bit[31]) This bit is always derived from the pin

Table 3-1. Power On Configuration Pin Assignments.

3.2.2 POC: Clock Configuration

The POC[3:1] bits enable the user to select from the pre-configured PLL settings in the table below. These frequencies are enabled during the power on process only, and the settings may be overridden by system software for operation.

POC[3:1]	VDSP (gclk_core)	IDSP (gclk_idsp)	DRAM (gclk_dram)	Vision (gclk_vision)	Cortex0 (gclk_cortex0)	Cortex1 (gclk_cortex1)
000	696	696	1824	792	456	1008
001	672	432	396	792	456	864
010	840	1008	2160	1320	552	1488
011	840	792	2160	744	456	1200
100	504	792	2160	1008	528	1320
101	504	408	1584	744	480	864
110	696	768	1920	720	480	1152
111	216	216	396	504	396	396

Table 3-2. Power On Clock Rates (MHz).

3.2.3 POC: Boot Mode Summary

Several bits of the power on configuration pins are decoded to determine the CV22FS boot mode. The following table shows the available boot-mode settings, including which pins must be controlled (with pull-up / pull-down) and which can be left unconnected (floating). Table 3-4 outlines the CV22FS shared boot options.

POC Bit	POC Description	Pin	SPI NOR Boot	SPI NAND Boot	MMC Boot	USB Boot	Boot Bypass (SPI NAND)	Boot Bypass (MMC)
0	Unused	**ENET_TXD_0**	0	0	0	0	0	0
3:1	clk_freq	**ENET_TXD_3, ENET_TXD_2, ENET_TXD_1**	0-7	0-7	0-7	0-7	0-7	0-7
4	boot_modes[0]	**ENET1_TXD_0**	0	1	0	x	1	0
5	boot_modes[1]	**ENET1_TXD_1**	0	0	1	x	0	1
6	secure_boot	**ENET1_TXD_2**	0/1	0/1	0/1	0/1	0/1	0/1
7	Reserved	-	-	-	-	-	-	-
8	boot_bypass	**CLK_AU**	x	0	0	x	1	1
9	Reserved	-	-	-	-	-	-	-

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POC Bit	POC Description	Pin	SPI NOR Boot	SPI NAND Boot	MMC Boot	USB Boot	Boot Bypass (SPI NAND)	Boot Bypass (MMC)
10	force_usb_boot	**ENET_PTP_PPS**	0	0	0	1	0	0
12:11	Reserved	-	-	-	-	-	-	-
13	boot_options[0]	**UART0_AHB_TX**	x	0/1	0/1	x	0/1	0/1
14	boot_options[1]	**UART1_AHB_TX**	0/1	0/1	0/1	x	0/1	0/1
15	boot_options[2]	**UART_APB_TX**	0/1	0/1	0/1	x	0/1	0/1
16	boot_options[3]	**I2S_SO_0**	0/1	0/1	0/1	x	0/1	0/1
17	boot_options[4]	**GPIO25**	0/1	0/1	0/1	x	0/1	0/1
18	boot_options[5]	**I2S_WS**	0/1	0/1	0/1	x	0/1	0/1
19	boot_options[6]	**SVSYNC**	x	x	0/1	x	x	0/1
20	boot_options[7]	**SHSYNC**	x	0/1	0/1	x	0/1	0/1
21	Unused	**ENET_GTX_CLK**	0	0	0	0	0	0
22	Reserved	-	-	-	-	-	-	-
23	clk_mode[0]	**OTP ONLY**	0/1	0/1	0/1	0/1	0/1	0/1
24	clk_mode[1]	**OTP ONLY**	0/1	0/1	0/1	0/1	0/1	0/1
25	Reserved	-	-	-	-	-	-	-
26	periph_clock_mode	**OTP ONLY**	0/1	0/1	0/1	0/1	0/1	0/1
29:27	Reserved	-	-	-	-	-	-	-
30	Unused	**PWM0**	0	0	0	0	0	0
31	Sysconfig-src	**PWM1**	0/1	0/1	0/1	0/1	0/1	0/1

Table 3-3. Power On Boot Mode.

Notes:

- Boot bypass is supported only for SPI NAND and eMMC boot, and is used for debugging purposes.
- The boot_options value will be ignored during boot bypass; however, this value cannot be unknown—it should be the same as the boot option for SPI NAND / MMC boot.
- Flash includes the following limitation: clk_core < 500 MHz. peripheral_clk_mode = 1 can only be set when clk_core is high due to a POC[3:1] setting.
- For MMC boot, use eMMC with a specification higher than or equal to Version 4.4, which supports boot function. A generic SD card or an eMMC 4.1 *should not* be used.
- For the hardware connection, use SD0 pins to connect to eMMC (4.4 or later).

Boot Option	SPI NOR	SPI NAND	eMMC	Force USB
boot_option[7]		8K BST code size select	emmc_bst_8k	
boot_option[6]			trsca_sel_boot	
boot_option[5]	Reset command	SCK mode	trsca_en_boot	
boot_option[4]	Use 2-byte address	4K-page select 2K-page select	emmc_reset_default_on	
boot_option[3]	Use alternative clock	Skip ECC check for blank pages	emmc_boot_sd_ pull	

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Boot Option	SPI NOR	SPI NAND	eMMC	Force USB
boot_option[2]	SPI mode selection	SPI NAND ECC BCH enable	sd4_boot	
boot_option[1]	LSB first	SPI NAND spare cell 2X	sd8_boot	
boot_option[0]		Flash fast boot		

Table 3-4. Shared Boot Options Matrix.

3.2.4 POC: Force USB Boot

The USB boot option is enabled using POC bit 10. Setting this pin to high overrides the current boot mode setting and forces the chip into USB boot mode.

3.2.5 POC: SPI NOR Boot

SPI NOR boot options are listed below. Five boot option pins are required.

Boot Option	Description	POC Pin
Reset command	Reset command for the external SPI NOR device 0 Send 0x66 - 0x99 1 Send 0xFF - 0xF0	18
2-byte address	Number of bytes in SPI NOR command address phase 0 3 bytes 1 2 bytes	17
Use alternative clock	SPI NOR reference clock selection 0 Use GCLK_ENET as SPI NOR controller reference clock 1 Use GCLK_CORE as SPI NOR controller reference clock	16
SPI mode selection	SPI mode selection 0 SCPOL = 0 (SPI mode 00) Always set to 0	15
LSB first	Controls TX / RX data swap 0 MSB first (do not swap TX send data and RX receive data) 1 LSB first (swap TX send data and RX receive data)	14

Table 3-5. SPI NOR Boot Options.

Note:

Boot code reads RCT system configuration register and configures SPI NOR.

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3.2.6 POC: SPI NAND Boot

SPI NAND boot options are listed below. Seven boot option pins are required.

Boot Option	Description	POC Pin
bst_8k	0 4K BST code size 1 8K BST code size	20
SCK mode	0 Mode 0 (CPOL = 0. CPHA = 0) 1 Mode 3 (CPOL = 1. CPHA = 1)	18
2k_pg	0 4K-page 1 2K-page	17
skip_bnk_ecc	0 Do not skip ECC check for blank pages 1 Skip ECC check for blank pages	16
ecc_bch	Enable ECC BCH function or not	15
sp_2x	0 Support 1X SPI NAND spare area When SPI NAND page size is 0: Spare area = 128 B When SPI NAND page size is 1: Spare area = 64 B 1 Support 2X SPI NAND spare area When SPI NAND page size is 0: Spare area = 256 B When SPI NAND page size is 1: Spare area = 128 B	14
flash_fastboot	0 Normal SPI NAND flash component read cycle time 1 Short SPI NAND flash component read cycle time	13

Table 3-6. SPI NAND Boot Options.

Note:

RCT system configuration register drives SPI NAND controller input.

3.2.7 POC: MMC Boot

MMC boot options are listed below. Seven boot option pins are required.

Boot Option	Description	POC Pin
emmc_bst_8k	0 4K BST code size 1 8K BST code size	20
trsca_sel_boot	0 For tRSCA=75 SD clock cycles 1 For tRSCA=10000 SD clock cycles	19
trsca_en_boot	Set this value to 1 to delay the timing of boot initialization	18
emmc_reset_default_on	When EMMC boot = 1 and emmc_reset_default_on = 1, the SD controller drives sd_reset signal SMIO_35	17
emmc_boot_sd_pull	0 No pull on sd_cmd, and sd_d[7:0] 1 Pull on sd_cmd, and sd_d[7:0]	16
sd4_boot	Use 4 SD data pins for booting	15

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Boot Option	Description	POC Pin
sd8_boot	Use 8 SD data pins for booting	14

Table 3-7. MMC Boot Options.

The following table lists the MMC boot options for pull enable.

Pad Name	Boot Mode Setting	boot_sd_pull setting	boot_data_bit setting
	boot_mode (POC[5:4])	emmc_boot_sd_pull (boot_option[3])	{sd4_boot, sd8_boot} (boot_option[2:1])
smio_3(sd_cmd)	2'b10	1'b1	2'b00(1-bit boot) 2'b10(4-bit boot) 2'bx1(8-bit boot)
smio_18(sd_d0)			2'b00(1-bit boot) 2'b10(4-bit boot) 2'bx1(8-bit boot)
smio_19(sd_d1)			2'b10(4-bit boot) 2'bx1(8-bit boot)
smio_20(sd_d2)			2'b10(4-bit boot) 2'bx1(8-bit boot)
smio_21(sd_d3)			2'b10(4-bit boot) 2'bx1(8-bit boot)
smio_22(sd_d4)			2'bx1(8-bit boot)
smio_23(sd_d5)			2'bx1(8-bit boot)
smio_24(sd_d6)			2'bx1(8-bit boot)
smio_25(sd_d7)			2'bx1(8-bit boot)

Table 3-8. MMC Boot Options for Pull Enable.

Notes:

- Pads pull_en and pull_sel are decided by boot_mode, emmc_boot_sd_pull, {sd4_boot, and sd8_boot}
- eMMC 1-bit boot sd_cmd and sd_d0 must pull up
- eMMC 4-bit boot sd_cmd, sd_d0, sd_d1, sd_d2, sd_d3, sd_d4 must pull up
- eMMC 8-bit boot sd_cmd, sd_d0, sd_d1, sd_d2, sd_d3, sd_d4, sd_d5, sd_d6, sd_d7 must pull up

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The following lists the MMC boot modes.

EMMC Boot Mode	Boot Mode Setting	boot_sd_pull setting	boot_data_bit setting
	boot_mode (POC[5:4])	emmc_boot_sd_pull (boot_option[3])	{sd4_boot, sd8_boot} (boot_option[2:1])
eMMC 1-bit boot	2'b10	1'b1	2'b00 (1-bit boot)
eMMC 4-bit boot			2'b10 (4-bit boot)
eMMC 8-bit boot			2'bx1 (8-bit boot)

Table 3-9. MMC Boot Mode.

Notes:

- The RCT system configuration register drives SD / SDIO controller input.
- When eMMC boot mode is chosen, the **SD_RESET** pin must be pulled low by the external resistor.

3.2.8 POC: Boot Bypass

The boot bypass option is enabled using POC[8]. Boot bypass can be used when the boot process from media (SPI NAND and eMMC) will be bypassed, and also to enable system debugging.

3.2.9 POC: OTP ROM

The CV22FS chip enables users to select which power on configuration bits are loaded from OTP ROM or from strapping pins.

The CV22FS OTP ROM region for POC function is 64 bits; i.e., the lower 32 bits store system configuration data and the upper 32 bits handle selection.

If strapping pin 31 (**PWM1**) is pulled down, all configuration values are loaded from the pins. If it is pulled up, all power on configuration bits are loaded from OTP, with the exception of POC bits 10 and 31, which are always loaded from POC pins.

The contents of the OTP ROM module are set at the factory and are not user-accessible.

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4. VIDEO INPUT

4.1 VIN: Overview

There are six VIN instances in the CV22FS VIN module–one main VIN and two secondary VINs, and three virtual channel only VINs. For further details on VIN features, refer to the CV22FS datasheet.

This chapter provides system hardware details for the CV22FS video input module, including pin connections for the sensors, power and IO levels, flash strobe control, sync, termination, and VIN interface disconnect.

The chapter is organized as follows:

Primary Sensor Connections:

- (Section 4.2) VIN: MIPI CSI-2 Interface

Power, IO, and Misc Details:

- (Section 4.3) VIN: Power and I/O Level
- (Section 4.4) VIN: Flash Strobe Control
- (Section 4.5) VIN: Sensor VSync Connection and Voltage Notice
- (Section 4.6) VIN: Termination
- (Section 4.7) VIN: Disconnect

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4.2 VIN: MIPI CSI-2 Interface

- VIN0 supports up to 8-lane MIPI.CSI-2 inputs.
- VIN1 and VIN8 support up to 4-lane MIPI.
- VIN2, VIN3, and VIN4 support up to 4-lane MIPI virtual channel.
- The MIPI CSI-2 interface requires no external termination.
- Unused input signals should be tied to ground.
- The sensor input pins are mapped for MIPI CSI-2 according to the table in Section 4.2.1.

4.2.1 VIN MIPI CSI-2 Mode: Pin Map

The following table provides the MIPI CSI-2 pin map.

Pin Name	MIPI	
	8+4	4+4+4
SPCLK_P_0	CLK0_P (VIN0)	CLK0_P (VIN0)
SPCLK_N_0	CLK0_N (VIN0)	CLK0_N (VIN0)
SPCLK_P_1		CLK0_P (VIN8)
SPCLK_N_1		CLK0_N (VIN8)
SPCLK_P_2	CLK0_P (VIN1)	CLK0_P (VIN1)
SPCLK_N_2	CLK0_N (VIN1)	CLK0_N (VIN1)
SD_LVDS_P_0	D0_P (VIN0)	D0_P (VIN0)
SD_LVDS_N_0	D0_N (VIN0)	D0_N (VIN0)
SD_LVDS_P_1	D1_P (VIN0)	D1_P (VIN0)
SD_LVDS_N_1	D1_N (VIN0)	D1_N (VIN0)
SD_LVDS_P_2	D2_P (VIN0)	D2_P (VIN0)
SD_LVDS_N_2	D2_N (VIN0)	D2_N (VIN0)
SD_LVDS_P_3	D3_P (VIN0)	D3_P (VIN0)
SD_LVDS_N_3	D3_N (VIN0)	D3_N (VIN0)
SD_LVDS_P_4	D4_P (VIN0)	D0_P (VIN8)
SD_LVDS_N_4	D4_N (VIN0)	D0_N (VIN8)
SD_LVDS_P_5	D5_P (VIN0)	D1_P (VIN8)
SD_LVDS_N_5	D5_N (VIN0)	D1_N (VIN8)
SD_LVDS_P_6	D6_P (VIN0)	D2_P (VIN8)
SD_LVDS_N_6	D6_N (VIN0)	D2_N (VIN8)
SD_LVDS_P_7	D7_P (VIN0)	D3_P (VIN8)
SD_LVDS_N_7	D7_N (VIN0)	D3_N (VIN8)
SD_LVDS_P_8	D0_P (VIN1)	D0_P (VIN1)
SD_LVDS_N_8	D0_N (VIN1)	D0_N (VIN1)
SD_LVDS_P_9	D1_P (VIN1)	D1_P (VIN1)
SD_LVDS_N_9	D1_N (VIN1)	D1_N (VIN1)
SD_LVDS_P_10	D2_P (VIN1)	D2_P (VIN1)
SD_LVDS_N_10	D2_N (VIN1)	D2_N (VIN1)
SD_LVDS_P_11	D3_P (VIN1)	D3_P (VIN1)

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Pin Name	MIPI	
	8+4	4+4+4
SD_LVDS_N_11	D3_N (VIN1)	D3_N (VIN1)

Table 4-1. MIPI CSI-2 Pin Map.

Note:

Data lanes are shared between VIN0, VIN2, VIN3 and VIN4 only.

4.3 VIN: Power and I/O Level

The following table lists the VIN sensor power and IO level.

Mode	Power Rail
	MIPI_AVDD18_IO
MIPI CSI	1.2 V

Table 4-2. VIN Power Rail.

Notes:

- Pay close attention to the signal level/direction between the sensor and the CV22FS chip.
- Note that CV22FS offers 1.0-V and 1.8-V video input modes and does not support 3.3-V sensor interface power. Refer to the CV22FS chip datasheet for operating range and additional electrical characteristics.

4.4 VIN: Flash Strobe Control

The CV22FS flash strobe is controlled using stepper control functions.

4.5 VIN: Sensor VSync Connection and Voltage Notice

Sensor operation modes and characteristics vary with sensor design. When configuring VSync signal connections (e.g., sensor vertical sync, frame sync, or frame valid) between the CV22FS chip and an associated sensor, please note that a high degree of care should be exercised.

4.6 VIN: Termination

The receiver (CV22FS side) of the CV22FS Sub-LVDS interface provides a built-in 100-Ohm termination. This eliminates the need to incorporate termination resistors in the external circuit during PCB design.

4.7 VIN: Disconnect

When running the CV22FS chip without the Video / Sensor interface:

- Input pins **SD_LVDS_P/N_[11:0]** and **SPCLK_LVDS_P/N_[2:0]** can be ganged and connected to ground if not used.
- Output pins **SHSYNC** and **SVSYNC** can be left unconnected (floating).

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5. VIDEO OUTPUT

5.1 VOUT: Overview

This chapter provides system hardware detail for the CV22FS video output (VOUT) interfaces. The chapter is organized as follows:

- (Section 5.2) VOUT: Interfaces

5.2 VOUT: Interfaces

The video output (VOUT) interface supports two digital MIPI video outputs. The output ports are on two logical video channels as shown in the diagram below.

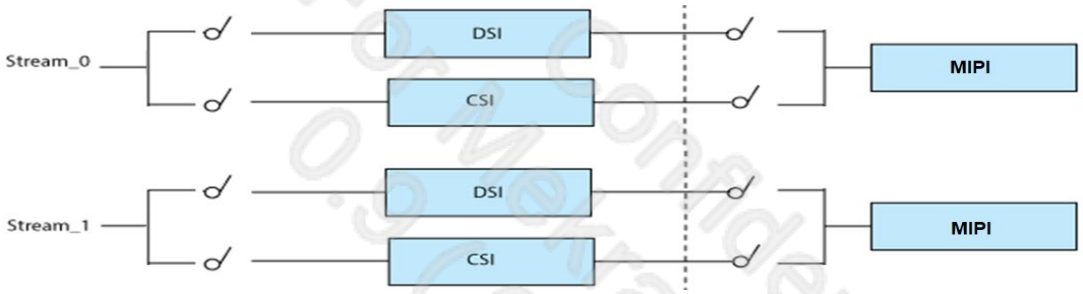


Figure 5-1. CV22FS Video Output Channels and Ports.

Refer to the CV22FS chip datasheet for information regarding the supported video output resolutions.

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6. GENERAL PURPOSE INPUT / OUTPUT PINS

6.1 GPIO: Overview

This chapter provides system hardware detail for CV22FS multi-function general purpose input / output (GPIO) pins. The chapter is organized as follows:

- (Section 6.2) GPIO: Characteristics
- (Section 6.3) GPIO: Function Selection
- (Section 6.4) GPIO: Alternations and Default States
- (Section 6.5) GPIO: SD Interface Pull High / Low Requirements
- (Section 6.6) GPIO: Disconnect
- (Section 6.7) GPIO: Implementing USB Host Function
- (Section 6.8) GPIO: Enhancing EIS Performance
- (Section 6.9) GPIO: Note about the CAN Bus
- (Section 6.10) GPIO: Note about the SPI NOR

6.2 GPIO: Characteristics

There are up to 103 GPIO pins included on CV22FS SoCs, each with multi-function capabilities, and with the following characteristics:

- CMOS type and programmable input / output
- Interrupt-capable
- Initialized to be inputs during reset assertion
- Bidirectional under software control

Note:

All CV22FS GPIOs do not support the fail-safe function.

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6.3 GPIO: Function Selection

The CV22FS GPIO pins have multiplexed functions for various hardware modules. Each GPIO has five alternation functions, and the IOMUX register set controls the alternation function selection. Refer to the *Ambarella CV22FS Hardware Programming Reference Manual* for details of IOMUX registers and programming procedures.

The boot device IO pins are part of the GPIO multiplexed functions. During system boot, the boot device is chosen by the POC[4] and POC[5] (boot mode select) pins; the two pins are also used to generate internal boot control signals “rct_ahb_*_boot”.

Table 6-1 shows POR_Alt columns that are mapped to the related alternation functions. When one of the boot control signals in the POR_Alt column is set, the corresponding alternation function is chosen for boot device pins; otherwise, GPIO function is chosen, by default. (Table 6-2 provides how to control these boot control signals.)

Note:

Refer to Chapter 3 in this document for the description of system boot and POC pins.

GPIO	Pin Name	POR_Alt1	POR_Alt2	POR_Alt3	POR_Alt4	POR_Alt5
0	**TIMER0**					
1	**IDCSCLK**					
2	**IDCSDATA**					
3	**SSI0CLK**	rct_ahb_ssi_clk_sel				
4	**SSI0MOSI**	rct_ahb_ssi_out_sel				
5	**SSI0MISO**	rct_ahb_ssi_in_sel				
6	**SSI0EN0**	rct_ahb_ssi_out_sel				
7	**SSI0EN1**	rct_ahb_ssi_other_sel				
8	**SSI2CLK**					
9	**SSI2MOSI**					
10	**SSI2MISO**					
11	**SSI2EN0**					
12	**SSI3CLK**			rct_ahb_ssis_clk_sel		
13	**SSI3MOSI**			rct_ahb_ssis_out_sel		
14	**SSI3MISO**			rct_ahb_ssis_in_sel		
15	**SSI3EN0**			rct_ahb_ssis_in_sel		
16	**UART_APB_RX**					
17	**UART_APB_TX**					
18	**PWM0**					
19	**PWM1**					
20	**PWM2**					
21	**I2S_CLK**	rct_ahb_i2s_sel				
22	**I2S_SI_0**	rct_ahb_i2s_sel				
23	**I2S_SO_0**	rct_ahb_i2s_sel				
24	**GPIO24**	rct_ahb_i2s_sel				

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GPIO	Pin Name	POR_Alt1	POR_Alt2	POR_Alt3	POR_Alt4	POR_Alt5
25	**GPIO25**	rct_ahb_i2s_sel				
26	**I2S_WS**	rct_ahb_i2s_sel				
-	**CLK_AU**					
27	**ENET_TXEN**	rct_ahb_rmii_tx_sel	rct_ahb_rgmii_ tx_sel			
28	**ENET_TXD_0**	rct_ahb_rmii_tx_sel	rct_ahb_rgmii_ tx_sel			
29	**ENET_TXD_1**	rct_ahb_rmii_tx_sel	rct_ahb_rgmii_ tx_sel			
30	**ENET_TXD_2**		rct_ahb_rgmii_ tx_sel			
31	**ENET_TXD_3**		rct_ahb_rgmii_ tx_sel			
32	**ENET_RXD_0**	rct_ahb_rmii_rx_ sel	rct_ahb_rgmii_ rx_sel			
33	**ENET_RXD_1**	rct_ahb_rmii_rx_ sel	rct_ahb_rgmii_ rx_sel			
34	**ENET_RXD_2**		rct_ahb_rgmii_ rx_sel			
35	**ENET_RXD_3**		rct_ahb_rgmii_ rx_sel			
36	**ENET_RXDV**	rct_ahb_rmii_rx_ sel	rct_ahb_rgmii_ rx_sel			
37	**ENET_MDC**	rct_ahb_rmii_oth- er_sel	rct_ahb_rgmii_ other_sel			
38	**ENET_MDIO**	rct_ahb_rmii_oth- er_sel	rct_ahb_rgmii_ other_sel			
39	**ENET_PTP_PPS**	rct_ahb_rmii_oth- er_sel	rct_ahb_rgmii_ other_sel			
40	**ENET_CLK_RX**	rct_ahb_rmii_clk_ sel	rct_ahb_rgmii_ rx_sel			
41	**ENET_GTX_CLK**	rct_ahb_rmii_oth- er_sel	rct_ahb_rgmii_ tx_sel			
42	**ENET_EXT_ OSC_CLK**	rct_ahb_rmii_oth- er_sel	rct_ahb_rgmii_ clk_sel			
43	**ENET_2ND_REF**	rct_ahb_rmii_oth- er_sel	rct_ahb_rgmii_ other_sel			
44	**ENET1_TXEN**	rct_ahb_rmii1_tx_ sel	rct_ahb_rgmii1_ tx_sel			rct_ahb_iochar_ canc_rx
45	**ENET1_TXD_0**	rct_ahb_rmii1_tx_ sel	rct_ahb_rgmii1_ tx_sel			
46	**ENET1_TXD_1**	rct_ahb_rmii1_tx_ sel	rct_ahb_rgmii1_ tx_sel			
47	**ENET1_TXD_2**		rct_ahb_rgmii1_ tx_sel			
48	**ENET1_TXD_3**		rct_ahb_rgmii1_ tx_sel			rct_ahb_iochar_ canc_tx
49	**ENET1_RXD_0**	rct_ahb_rmii1_rx_ sel	rct_ahb_rgmii1_ rx_sel			

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GPIO	Pin Name	POR_Alt1	POR_Alt2	POR_Alt3	POR_Alt4	POR_Alt5
50	**ENET1_RXD_1**	rct_ahb_rmii1_rx_sel	rct_ahb_rgmii1_rx_sel			
51	**ENET1_RXD_2**		rct_ahb_rgmii1_rx_sel			
52	**ENET1_RXD_3**		rct_ahb_rgmii1_rx_sel			
53	**ENET1_RXDV**	rct_ahb_rmii1_rx_sel	rct_ahb_rgmii1_rx_sel			rct_ahb_iochar_canc_rx
54	**ENET1_MDC**	rct_ahb_rmii1_other_sel	rct_ahb_rgmii1_other_sel			rct_ahb_iochar_canc_tx
55	**ENET1_MDIO**	rct_ahb_rmii1_other_sel	rct_ahb_rgmii1_other_sel			rct_ahb_iochar_canc_rx
56	**ENET1_PTP_PPS**	rct_ahb_rmii1_other_sel	rct_ahb_rgmii1_other_sel			rct_ahb_iochar_canc_tx
57	**ENET1_CLK_RX**	rct_ahb_rmii1_clk_sel	rct_ahb_rgmii1_rx_sel			rct_ahb_iochar_canc_rx
58	**ENET1_GTX_CLK**	rct_ahb_rmii1_other_sel	rct_ahb_rgmii1_tx_sel			rct_ahb_iochar_canc_tx
-	**JTAG_CLK**					
-	**JTAG_RST_L**					
-	**JTAG_TDI**					
-	**JTAG_TDO**					
-	**JTAG_TMS**					
-	**DETECT_VBUS**					
59	**SMIO_0**		rct_ahb_snand_boot	rct_ahb_nor-spi_boot		
60	**SMIO_1**		rct_ahb_snand_boot	rct_ahb_nor-spi_boot		
61	**SMIO_2**		rct_ahb_mmc_sd_boot			
62	**SMIO_3**		rct_ahb_mmc_sd_boot			
63	**SMIO_4**		rct_ahb_mmc_sd_boot			
64	**SMIO_5**		rct_ahb_mmc_sd_boot			
65	**SMIO_6**		rct_ahb_snand_boot	rct_ahb_nor-spi_boot		
66	**SMIO_7**		rct_ahb_snand_boot	rct_ahb_nor-spi_boot		
67	**SMIO_8**		rct_ahb_snand_boot	rct_ahb_nor-spi_boot		
68	**SMIO_9**		rct_ahb_snand_boot	rct_ahb_nor-spi_boot		
69	**SMIO_18**		rct_ahb_mmc_sd_boot			
70	**SMIO_19**		rct_ahb_mmc_sd_boot			
71	**SMIO_20**		rct_ahb_mmc_sd_boot			

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GPIO	Pin Name	POR_Alt1	POR_Alt2	POR_Alt3	POR_Alt4	POR_Alt5
72	**SMIO_21**		rct_ahb_mmc_sd_boot			
73	**SMIO_22**		rct_ahb_mmc_sd_boot	rct_ahb_nor-spi_boot		
74	**SMIO_23**		rct_ahb_mmc_sd_boot	rct_ahb_nor-spi_boot		
75	**SMIO_24**		rct_ahb_mmc_sd_boot	rct_ahb_nor-spi_boot		
76	**SMIO_25**		rct_ahb_mmc_sd_boot	rct_ahb_nor-spi_boot		
77	**SMIO_26**		rct_ahb_iochar_sdio			
78	**SMIO_27**		rct_ahb_iochar_sdio			
79	**SMIO_28**		rct_ahb_iochar_sdio			
80	**SMIO_29**		rct_ahb_iochar_sdio			
81	**SMIO_30**		rct_ahb_iochar_sdio			
82	**SMIO_31**		rct_ahb_iochar_sdio			
83	**SMIO_32**		rct_ahb_iochar_sdio			
84	**SMIO_33**		rct_ahb_iochar_sdio			
85	**SMIO_34**			rct_ahb_nor-spi_boot		
86	**SMIO_35**		rct_ahb_mmc_sd_reset			
87	**SENSOR_RST**					
-	**CLK_SI**					
-	**CLK_SI_1**					
88	**SHSYNC**					
89	**SVSYNC**					
90	**UART0_AHB_RX**					
91	**UART0_AHB_TX**					
92	**UART0_AHB_CTS_N**					
93	**UART0_AHB_RTS_N**					
94	**UART1_AHB_RX**					
95	**UART1_AHB_TX**					
96	**UART1_AHB_CTS_N**					
97	**UART1_AHB_RTS_N**					

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GPIO	Pin Name	POR_Alt1	POR_Alt2	POR_Alt3	POR_Alt4	POR_Alt5
98	**CAN0_RX**	rct_ahb_iochar_canc_rx				
99	**CAN0_TX**	rct_ahb_iochar_canc_tx				
100	**CAN1_RX**	rct_ahb_iochar_canc_rx				
101	**CAN1_TX**	rct_ahb_iochar_canc_tx				
102	**SD_HS_DEL**					
-	**CEHU_SAFETY_ERR_L_0**					
-	**CEHU_SAFETY_ERR_L_1**					
-	**TEST_MODE**					
-	**POR_L**					

Table 6-1. Alternation Functions for Boot Device GPIO Pins.

6.3.1 Controlling the Boot Control Signals

The following table lists how to control the boot control signals listed in the POR_Alt columns Table 6-1.

Signal Name in POR_Alt	How to Control	Notes
rct_ahb_norspi_boot	POC[5:4] 00: 1 Others: 0	SPI NOR boot selection.
rct_ahb_snand_boot	POC[5:4] 01: 1 Others: 0	SPI-NAND boot selection.
rct_ahb_mmc_sd_boot	POC[5:4] 10: 1 Others: 0	MMC boot selection.
rct_ahb_mmc_sd_reset	POC[17, 5:4] 110: 1 Others: 0	MMC reset selection for SD device to drive MMC reset at MMC boot.
rct_ahb_rmii_clk_sel	Control registers in RCT	ENET function selection with RMII mode
rct_ahb_rmii_tx_sel		
rct_ahb_rmii_rx_sel		
rct_ahb_rmii_other_sel		
rct_ahb_rgmii_clk_sel	Control registers in RCT	ENET function selection with RGMII mode
rct_ahb_rgmii_tx_sel		
rct_ahb_rgmii_rx_sel		
rct_ahb_rgmii_other_sel		
rct_ahb_rmii1_clk_sel	Control registers in RCT	ENET1 function selection with RMII mode.
rct_ahb_rmii1_tx_sel		
rct_ahb_rmii1_rx_sel		
rct_ahb_rmii1_other_sel		

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Signal Name in POR_Alt	How to Control	Notes
rct_ahb_rgmii1_clk_sel	Control registers in RCT.	ENET1 function selection with RGMII mode.
rct_ahb_rgmii1_tx_sel		
rct_ahb_rgmii1_rx_sel		
rct_ahb_rgmii1_other_sel		
rct_ahb_ssi_*	These signals are always tied off in functional mode.	
rct_ahb_ssis_*		
rct_ahb_i2s_*		
rct_ahb_iochar_*		

Table 6-2. Controlling the Boot Control Signals.

6.4 GPIO: Alternations and Default States

The table below provides the multiplexed functions, related POR alternation selection, and the state of pins during/after POR that contains default input / output direction and pull high / low characteristics for the CV22FS GPIO pins.

GPIO	Pin Name	Primary Function	Power Group	State During POR	State After POR
0	**TIMER0**	TM11_CLK	GPIO_IO	No pull input	No pull input
1	**IDCSCLK**	IDCSCLK	GPIO_IO	No pull input	No pull input
2	**IDCSDATA**	IDCSDATA	GPIO_IO	No pull input	No pull input
3	**SSI0CLK**	SSI0_SCLK	GPIO_IO	No pull input	No pull input
4	**SSI0MOSI**	SSI0_TXD	GPIO_IO	No pull input	No pull input
5	**SSI0MISO**	SSI0_RXD	GPIO_IO	No pull input	No pull input
6	**SSI0EN0**	SSI0_EN0	GPIO_IO	No pull input	No pull input
7	**SSI0EN1**	SSI0_EN1	GPIO_IO	No pull input	No pull input
8	**SSI2CLK**	SSI2_SCLK	GPIO_IO	No pull input	No pull input
9	**SSI2MOSI**	SSI2_TXD	GPIO_IO	No pull input	No pull input
10	**SSI2MISO**	SSI2_RXD	GPIO_IO	No pull input	No pull input
11	**SSI2EN0**	SSI2_EN0	GPIO_IO	No pull input	No pull input
12	**SSI3CLK**	SSI3_SCLK	GPIO_IO	No pull input	No pull input
13	**SSI3MOSI**	SSI3_TXD	GPIO_IO	No pull input	No pull input
14	**SSI3MISO**	SSI3_RXD	GPIO_IO	No pull input	No pull input
15	**SSI3EN0**	SSI3_EN0	GPIO_IO	No pull input	No pull input
16	**UART_APB_RX**	UART_APB_RX	GPIO_IO	No pull input	No pull input
17	**UART_APB_TX**	UART_APB_TX	GPIO_IO	No pull input	No pull input
18	**PWM0**	PWM0	GPIO_IO	No pull input	No pull input
19	**PWM1**	PWM1	GPIO_IO	No pull input	No pull input
20	**PWM2**	PWM2	GPIO_IO	No pull input	No pull input
21	**I2S_CLK**	I2S_CLK	GPIO_IO	No pull input	No pull input
22	**I2S_SI_0**	I2S_SI_0	GPIO_IO	No pull input	No pull input
23	**I2S_SO_0**	I2S_SO_0	GPIO_IO	No pull input	No pull input
24	**GPIO24**		GPIO_IO	No pull input	No pull input
25	**GPIO25**		GPIO_IO	No pull input	No pull input
26	**I2S_WS**	I2S_WS	GPIO_IO	No pull input	No pull input
-	**CLK_AU**	CLK_AU		No pull input	No pull input

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GPIO	Pin Name	Primary Function	Power Group	State During POR	State After POR
27	**ENET_TXEN**	ENET_TXEN	GPIO_IO	No pull input	No pull input
28	**ENET_TXD_0**	ENET_TXD_0	GPIO_IO	No pull input	No pull input
29	**ENET_TXD_1**	ENET_TXD_1	GPIO_IO	No pull input	No pull input
30	**ENET_TXD_2**		GPIO_IO	No pull input	No pull input
31	**ENET_TXD_3**		GPIO_IO	No pull input	No pull input
32	**ENET_RXD_0**	ENET_RXD_0	GPIO_IO	No pull input	No pull input
33	**ENET_RXD_1**	ENET_RXD_1	GPIO_IO	No pull input	No pull input
34	**ENET_RXD_2**		GPIO_IO	No pull input	No pull input
35	**ENET_RXD_3**		GPIO_IO	No pull input	No pull input
36	**ENET_RXDV**	ENET_RXDV	GPIO_IO	No pull input	No pull input
37	**ENET_MDC**	ENET_MDC	GPIO_IO	No pull input	No pull input
38	**ENET_MDIO**	ENET_MDIO	GPIO_IO	No pull input	No pull input
39	**ENET_PTP_PPS**	ENET_PTP_ PPS_O	GPIO_IO	No pull input	No pull input
40	**ENET_CLK_RX**	ENET_REF_CLK	GPIO_IO	No pull input	No pull input
41	**ENET_GTX_CLK**	ENET_2ND_ REF_CLK	GPIO_IO	No pull input	No pull input
42	**ENET_EXT_ OSC_ CLK**	ENET_EXT_ OSC_CLK	GPIO_IO	No pull input	No pull input
43	**ENET_2ND_REF**	ENET_2ND_ REF_CLK	GPIO_IO	No pull input	No pull input
44	**ENET1_TXEN**	ENET1_TXEN	GPIO_IO	No pull input	No pull input
45	**ENET1_TXD_0**	ENET1_TXD_0	GPIO_IO	No pull input	No pull input
46	**ENET1_TXD_1**	ENET1_TXD_1	GPIO_IO	No pull input	No pull input
47	**ENET1_TXD_2**		GPIO_IO	No pull input	No pull input
48	**ENET1_TXD_3**		GPIO_IO	No pull input	No pull input
49	**ENET1_RXD_0**	ENET1_RXD_0	GPIO_IO	No pull input	No pull input
50	**ENET1_RXD_1**	ENET1_RXD_1	GPIO_IO	No pull input	No pull input
51	**ENET1_RXD_2**	IDC5CLK	GPIO_IO	No pull input	No pull input
52	**ENET1_RXD_3**	IDC5DATA	GPIO_IO	No pull input	No pull input
53	**ENET1_RXDV**	ENET1_RXDV	GPIO_IO	No pull input	No pull input
54	**ENET1_MDC**	ENET1_MDC	GPIO_IO	No pull input	No pull input
55	**ENET1_MDIO**	ENET1_MDIO	GPIO_IO	No pull input	No pull input
56	**ENET1_PTP_ PPS**	ENET1_PTP_ PPS_O	GPIO_IO	No pull input	No pull input
57	**ENET1_CLK_RX**	ENET1_REF_ CLK	GPIO_IO	No pull input	No pull input
58	**ENET1_GTX_ CLK**	ENET_2ND_ REF_CLK	GPIO_IO	No pull input	No pull input
-	**JTAG_CLK**			No pull input	No pull input
-	**JTAG_RST_L**			No pull input	No pull input
-	**JTAG_TDI**			No pull input	No pull input
-	**JTAG_TDO**			No pull output	No pull output
-	**JTAG_TMS**			No pull input	No pull input
-	**DETECT_VBUS**			No pull input	No pull input
59	**SMIO_0**		NAND_IO	No pull input	Boot mode dependent
60	**SMIO_1**		NAND_IO	No pull input	Boot mode dependent
61	**SMIO_2**		SD_IO	No pull input	Boot mode dependent
62	**SMIO_3**		SD_IO	No pull input	Boot mode dependent
63	**SMIO_4**		SD_IO	No pull input	Boot mode dependent
64	**SMIO_5**		SD_IO	No pull input	Boot mode dependent
65	**SMIO_6**		NAND_IO	No pull input	Boot mode dependent
66	**SMIO_7**		NAND_IO	No pull input	Boot mode dependent

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GPIO	Pin Name	Primary Function	Power Group	State During POR	State After POR
67	**SMIO_8**		NAND_IO	No pull input	Boot mode dependent
68	**SMIO_9**		NAND_IO	No pull input	Boot mode dependent
69	**SMIO_18**		SD_IO	No pull input	Boot mode dependent
70	**SMIO_19**		SD_IO	No pull input	Boot mode dependent
71	**SMIO_20**		SD_IO	No pull input	Boot mode dependent
72	**SMIO_21**		SD_IO	No pull input	Boot mode dependent
73	**SMIO_22**		SD_IO	No pull input	Boot mode dependent
74	**SMIO_23**		SD_IO	No pull input	Boot mode dependent
75	**SMIO_24**		SD_IO	No pull input	Boot mode dependent
76	**SMIO_25**		SD_IO	No pull input	Boot mode dependent
77	**SMIO_26**	UART2_AHB_RX	SDIO0_IO	No pull input	Boot mode dependent
78	**SMIO_27**	UART2_AHB_TX	SDIO_IO	No pull input	Boot mode dependent
79	**SMIO_28**	UART2_AHB_ CTS_N	SDIO_IO	No pull input	Boot mode dependent
80	**SMIO_29**	UART2_AHB_ RTS_N	SDIO_IO	No pull input	Boot mode dependent
81	**SMIO_30**	UART3_AHB_ CTS_N	SDIO_IO	No pull input	Boot mode dependent
82	**SMIO_31**	UART3_AHB_ RTS_N	SDIO_IO	No pull input	Boot mode dependent
83	**SMIO_32**	UART3_AHB_RX	SDIO_IO	No pull input	Boot mode dependent
84	**SMIO_33**	UART3_AHB_TX	SDIO_IO	No pull input	Boot mode dependent
85	**SMIO_34**		NAND_IO	No pull input	Boot mode dependent
86	**SMIO_35**		SD_IO	No pull input	Boot mode dependent
87	**SENSOR_RST**		GPIO_IO	No pull input	No pull input
-	**CLK_SI**			No pull input	No pull input
-	**CLK_SI_1**			No pull input	No pull output
88	**SHSYNC**	IDSP_VIN0_IOPAD_ MASTER_ HSYNC	GPIO_IO	No pull input	No pull input
89	**SVSYNC**	IDSP_VIN0_IOPAD_ MASTER_ VSYNC	GPIO_IO	No pull input	No pull input
90	**UART0_AHB_RX**	UART0_AHB_RX	GPIO_IO	No pull input	No pull input
91	**UART0_AHB_TX**	UART0_AHB_TX	GPIO_IO	No pull input	No pull input
92	**UART0_AHB_ CTS_N**	UART0_AHB_ CTS_N	GPIO_IO	No pull input	No pull input
93	**UART0_AHB_ RTS_N**	UART0_AHB_ RTS_N	GPIO_IO	No pull input	No pull input
94	**UART1_AHB_RX**	UART1_AHB_RX	GPIO_IO	No pull input	No pull input
95	**UART1_AHB_TX**	UART1_AHB_TX	GPIO_IO	No pull input	No pull input
96	**UART1_AHB_ CTS_N**	UART1_AHB_ CTS_N	GPIO_IO	No pull input	No pull input
97	**UART1_AHB_ RTS_N**	UART1_AHB_ RTS_N	GPIO_IO	No pull input	No pull input
98	**CAN0_RX**	CAN0_RX	GPIO_IO	No pull input	No pull input
99	**CAN0_TX**	CAN0_TX	GPIO_IO	No pull input	No pull input
100	**CAN1_RX**	CAN1_RX	GPIO_IO	No pull input	No pull input
101	**CAN1_TX**	CAN1_TX	GPIO_IO	No pull input	No pull input
102	**SD_HS_DEL**	SD_HS_SEL	GPIO_IO	No pull input	No pull input
-	**CEHU_SAFETY_ ERR_L_0**			No pull input	No pull input
-	**CEHU_SAFETY_ ERR_L_1**			No pull input	No pull input

GPIO	Pin Name	Primary Function	Power Group	State During POR	State After POR
-	**TEST_MODE**			No pull input	No pull input
-	**POR_L**			No pull input	No pull input

Table 6-3. General Purpose Input Output (GPIO) Pin Default Characteristics.

6.4.1 Non-GPIO Power Groups

The following table lists the non-GPIO pins and their power groups.

Pin Name	Power Group
JTAG_CLK	GPIO_IO
JTAG_RST_L	GPIO_IO
JTAG_TDI	GPIO_IO
JTAG_TDO	GPIO_IO
JTAG_TMS	GPIO_IO
DETECT_VBUS	GPIO_IO
TEST_MODE	GPIO_IO
CLK_AU	GPIO_IO
CLK_SI	GPIO_IO
CLK_SI_1	GPIO_IO
CEHU_SAFETY_ERR_L_0	GPIO_IO
CEHU_SAFETY_ERR_L_1	GPIO_IO
POR_L	GPIO_IO

Table 6-4. Non-GPIO Pins and Power Groups.

6.4.2 EMMC Pull High / Low Characteristics

The following table lists the EMMC pull high / low characteristics.

Pin Name	Primary Function	EMMC 1-bit / 4-bit / 8-bit boot and POC EMMC_ BOOT_SD_PULL is NOT set	EMMC 1-bit boot and POC EMMC_ BOOT_SD_PULL is set	EMMC 4-bit boot and POC EMMC_ BOOT_SD_PULL is set	EMMC 8-bit boot and POC EMMC_ BOOT_SD_PULL is set
SMIO_3	SD_CMD	No pull	Pull high	Pull high	Pull high
SMIO_18	SD_D[0]	No pull	Pull high	Pull high	Pull high
SMIO_19	SD_D[1]	No pull	No pull	Pull high	Pull high
SMIO_20	SD_D[2]	No pull	No pull	Pull high	Pull high
SMIO_21	SD_D[3]	No pull	No pull	Pull high	Pull high
SMIO_22	SD_D[4]	No pull	No pull	No pull	Pull high
SMIO_23	SD_D[5]	No pull	No pull	No pull	Pull high
SMIO_24	SD_D[6]	No pull	No pull	No pull	Pull high
SMIO_25	SD_D[7]	No pull	No pull	No pull	Pull high

*Table 6-5. EMMC Pull High / Low Characteristics.**

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6.4.3 IO DC Specifications

The following lists the IO DC specifications for CV22FS.

Parameter	Symbol	Test Conditions	Min	Max	Unit
Input Pull-Up Resistor Current/Resistance	I_{RPU}	V_{PAD} = 0V	–20	–180	µA
	R_{RPU}	V_{PAD} = 0V	10	100	KΩ
Input Pull-Down Resistor Current/Resistance	I_{RPU}	V_{PAD} = VDDO	20	180	µA
	R_{RPU}	V_{PAD} = VDDO	10	100	KΩ

Table 6-6. DC Electrical Specifications.

6.5 GPIO: SD Interface Pull High / Low Requirements

SD Interface Type	SD Card	WiFi	eMMC	Unused
sd / sdio_cd	Pull high	Floating	Pull low	Pull high
sd / sdio_wp	Depends on SD socket structure	Floating	Pull low	Pull high

Table 6-7. GPIO: SD Interface Pull High / Low Control.

Notes:

- The pull high / low state should be controlled by external resistor.
- Note that when the SD0 interface is connected to an SD card slot, the unused SD0 interface data pins cannot function as GPIO pins.

6.6 GPIO: Disconnect

Multi-function GPIO pins are used to enable various interfaces and functions on the CV22FS chip including video output, UART, interval timers, and more. Selection, control, and disconnection of these functions is performed via software.

6.7 GPIO: Implementing USB Host Function

To implement the USB host function, two GPIOs must be programmed to the hardware function (overcurrent protection) and pulled low. There are five sets of GPIOs to implement this function:

- GPIO1 and GPIO2 with Alternative Function 5
- GPIO16 and GPIO17 with Alternative Function 2
- GPIO24 and GPIO25 with Alternative Function 3
- GPIO57 and GPIO58 with Alternative Function 4
- GPIO90 and GPIO92 with Alternative Function 3

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6.8 GPIO: Enhancing EIS Performance

In order to achieve video stabilization, sensor data and gyro data synchronization occurs in the EIS algorithm. PWM0/1/2/3 is a feasible candidate for the synchronization signal.

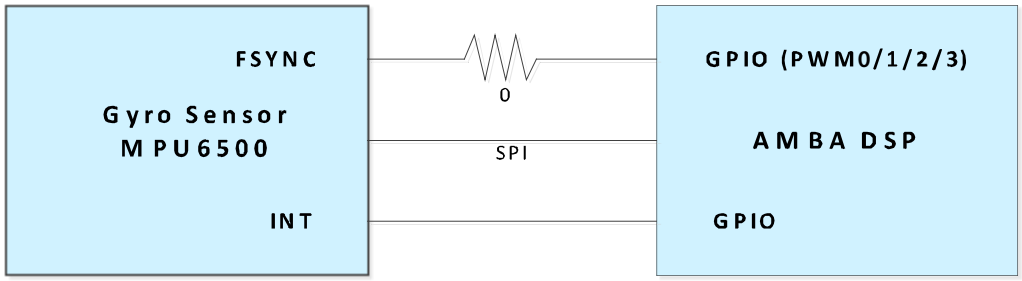


Figure 6-1. Sensor Data and Gyro Data Synchronization.

6.8.1 Gyro Sensor Mounting

Gyro placement suggestions include the following:

- The gyro sensor should be placed as close as possible to the image sensor to ensure that the gyro sensor detects the accurate image sensor movement required for EIS correction. However, because the gyro sensor characteristics can be affected by temperature, the heat from the image sensor is also an important consideration.
- The gyro sensor must be placed in the correct orientation and location. The following figure shows a mounting example for a three-axis gyro sensor, which should be oriented in parallel with the image sensor and placed at the locations indicated by the blue boxes in order to detect the correct image sensor movement.

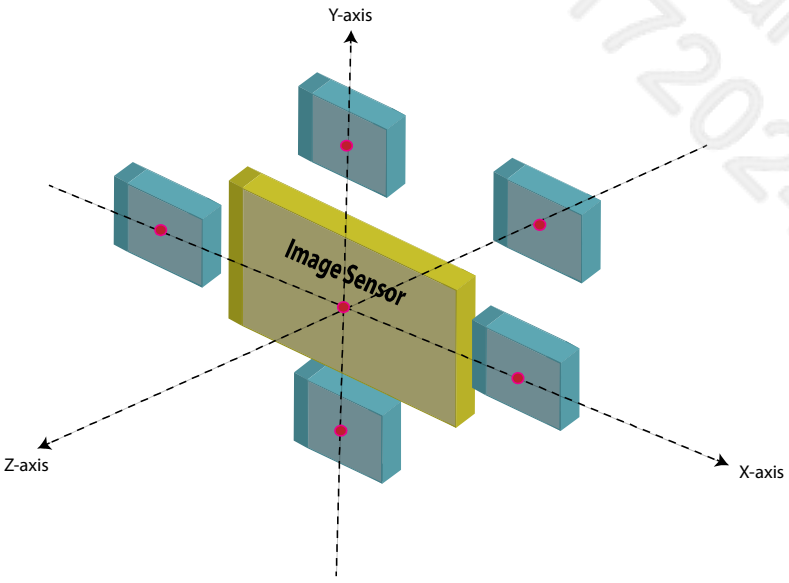


Figure 6-2. Three-Axis Gyro Sensor Mounting.

Note:

The image sensor and gyro sensor must have a direct mechanical coupling to ensure that both sensors experience exactly the same vibration; otherwise, EIS performance may be drastically degraded.

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6.9 GPIO: Note about the CAN Bus

All CAN nodes on the bus must be in *one* of two modes:

- Classic CAN, or
- CAN FD
  - No mixed format
  - All CAN FD packets' extended data length (EDL) fields should be set to 1.

6.10 GPIO: Note about the SPI NOR

When using an Octa SPI NOR device with eight I/O signals and a data strobe (DQS pin), the 8-lane DTR mode is not supported.

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7. ETHERNET

7.1 Ethernet: Overview

This chapter summarizes the CV22FS Ethernet interface as follows:

- (Section 7.2) Ethernet: Pins
- (Section 7.3) Ethernet: Disconnect

7.2 Ethernet: Pins

ENET_CLK_RX is input only when RMII is active. The Ethernet PHY side must be configured to output.

Please refer to the CV22FS chip datasheet for complete information regarding CV22FS Ethernet pin functionality.

7.3 Ethernet: Disconnect

The CV22FS Ethernet interface uses multi-function GPIO pins. Selection, control, and disconnection of these functions are performed via software.

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8. I2C / IDC PORTS

8.1 I2C / IDC: Overview

This chapter summarizes the CV22FS I2C / IDC ports as follows:

- (Section 8.2) I2C / IDC Ports

8.2 I2C / IDC Ports

The I2C / IDC interfaces control a two-wire, bi-directional bus that provides data communication between the chip and peripheral devices with protocol speeds up to 3.4 Mbps. The chip includes a maximum of six I2C / IDC ports. All of the CV22FS I2C / IDC interfaces support single-master mode.

Note that the timing margin for I2C / IDC interfaces 2 and 3 is critical. When supporting multiple I2C / IDC devices, a 100-pF capacitor must be added to the **IDC_CLK** line to delay the clock.

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9. JTAG AND UART INTERFACES

9.1 JTAG and UART: Overview

This chapter covers the CV22FS JTAG and UART interfaces as follows:

- (Section 9.2) JTAG and UART: JTAG
- (Section 9.3) JTAG and UART: UART
- (Section 9.4) JTAG and UART: Disconnect

9.2 JTAG and UART: JTAG

The CV22FS chip includes an interface for a JTAG in-circuit emulator (ICE). This is primarily used for debugging purposes. Information regarding the JTAG interface is provided below.

- JTAG chaining is not supported.
- The **JTAG_CLK** pin should be pulled low.
- The reset pin **JTAG_RST_L** should be pulled low during the power on period.
- **JTAG_TDI** / **JTAG_TMS** should be pulled high to save power (and to avoid power leakage).

9.3 JTAG and UART: UART

The CV22FS chip includes five industry-standard 16550 UART programming interfaces (UART_APB, UART0_AHB ~ UART3_AHB).

- UART0_AHB ~ UART3_AHB supports hardware flow control and may be used to communicate with external devices.
- The UART_APB interface is generally used for debugging purposes and does not provide hardware flow control.
  - The software transmits debug-related information through UART_APB for print-out on the serial console. To increase system robustness after the debug board is removed, pull-up or pull-down the Tx / Rx signals to eliminate unwanted noise/crosstalk (and unexpected interrupts).
  - The UART_APB interrupt should be disabled after system development is completed. While this eliminates debugging output through UART_APB, it increases system robustness and is therefore recommended.

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9.4 JTAG and UART: Disconnect

9.4.1 Disconnect: JTAG

When running the CV22FS chip without the JTAG interface:

- Pull-up the **JTAG_TMS** and **JTAG_TDI** input pins.
- Pull-down the **JTAG_CLK** and **JTAG_RST_L** input pins.
- Leave the output pin **JTAG_TDO** unconnected (floating).

9.4.2 Disconnect: UART

The UART interfaces are enabled with multi-function GPIO pins. Selection, control, and disconnection of these functions is performed via software.

- Individual GPIO functions are selected via register programming.
- The GPIO pins are individually pulled up/down using register programming.

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10. PCB LAYOUT REFERENCE

10.1 PCB Layout: Overview

This chapter provides CV22FS PCB layout information as a reference to be considered during board development. Note that actual PCB layouts are strongly application-dependent, and designs should be based on a variety of factors including PCB stack-up parameters and SI / PI simulations.

Also note that Ambarella provides qualified customers with hardware reference designs to aid in board layout and development, as well as design review services. It is expected that users will both refer to Ambarella-provided reference designs and consult directly with an Ambarella representative during product development efforts.

The chapter is organized as follows:

- (Section 10.2) PCB Layout: System ESD Protection Mechanism
- (Section 10.3) PCB Layout: DRAM
- (Section 10.4) PCB Layout: Sensor / Video Input
- (Section 10.5) PCB Layout: Ethernet
- (Section 10.6) PCB Layout: SMIO
- (Section 10.7) PCB Layout: USB
- (Section 10.8) PCB Layout: Oscillator
- (Section 10.9) PCB Layout: Crystal
- (Section 10.10) PCB Layout: Power Rails
- (Section 10.11) PCB Layout: Miscellaneous

10.2 PCB Layout: System ESD Protection Mechanism

CV22FS includes ESD diodes in the signal and power / ground IOs which meet the JEDEC and AQC-Q100 specification of 2000 V HBM and 500 V CDM. If the system requires a higher level of ESD, then external ESD protection devices are required at the PCB level. Users must ensure that the voltage level of the devices meets the IO level, and that the parasitics of the devices is low enough to not degrade signal integrity.

10.3 PCB Layout: DRAM

PCB layout guidelines for DRAM are provided in this section as follows:

- (Section 10.3.1) DRAM: Placement
- (Section 10.3.2) DRAM: Standard Guidelines
- (Section 10.3.3) DRAM: Decoupling Capacitors
- (Section 10.3.4) DRAM: Signal Routing Example for LPDDR4 DRAM
- (Section 10.3.5) DRAM: Internal Length

Contact an Ambarella representative for assistance in selecting a qualified DRAM component.

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10.3.1 DRAM: Placement

The following is a diagram roughly illustrating the recommended placement of DRAM modules relative to the CV22FS SoC. Please contact an Ambarella representative for further details.

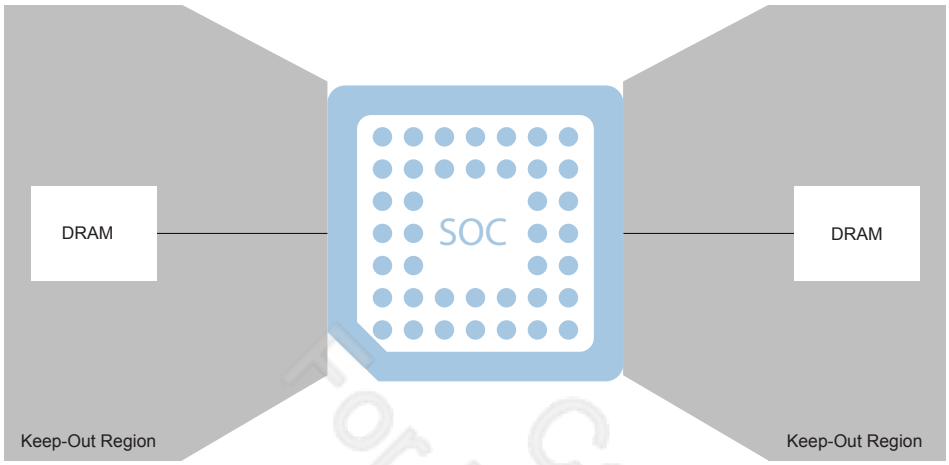


Figure 10-1. PCB Board Layout: DRAM Placement Recommendation.

Note*:*

DRAM modules should be placed as close to the CV22FS SoC as possible, given the guidelines below.

10.3.2 DRAM: Standard Guidelines

This section contains a list of commonly accepted guidelines for DRAM placement on a PCB board.

- For the ground reference plane, there should be a full, contiguous ground plane next to every DRAM routing layer.
- The keep-out region, which varies with individual design, is used to prevent other signals from interfering with the DRAM interface. The only signals allowed in the region (i.e., on the DRAM signal layers) are those for this interface. Do not route DRAM-irrelative traces in the DRAM keep-out region.
- Split reference planes are not permitted in the DRAM keep-out region. The 1.1-V power partial plane should encompass at least the entire DRAM region.
- The power supply rails for the DRAM device and the CV22FS chip should be implemented with minimum power-ground loop and the lowest possible degree of interference.
- Trace lengths should be set as short as possible to improve signal integrity.

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10.3.3 DRAM: Decoupling Capacitors

- Decoupling capacitors should be placed near the device being decoupled.
- The distance from the capacitor to the power pins being decoupled should not exceed 125 mils.
- Each decoupling capacitor requires two vias (i.e., one for each pin). Via-sharing for decoupling capacitors is not permitted.
- One decoupling capacitor per power / ground pin pair should be added whenever possible.

10.3.4 DRAM: Signal Routing Example for LPDDR4 DRAM

The following table lists the LPDDR4 routing recommendations.

Notes:

- Include SoC internal length.
- In LPDDR4 training mode, LPDDR4 DQ bits cannot be swapped.

Signal	Group	PCB + Package Propagation Delay		Considerations	Convert to Length (Assume Propagation Delay of 160 ps / in)
		Minimum	Maximum		
CK_P / CK_N	Clock	Short as possible	160 ps	[P&N] with +/- 5 mils	< 1000 mils
CA CS CKE	Address / Command / Control	CK_P - 16 ps	CK_P + 16 ps		CK_P +/- 100 mils
DQS0_P / DQS0_N	Byte0 DQS	CK_P - 40 ps	CK_P + 40 ps	[P&N] with +/- 5 mils	Within CK_P +/- 250 mils
DM0 DQ[7:0]	Byte0 Data	DQS0P - 8 ps	DQS0P + 8 ps		DQ / DM should be in reference to the corresponding DQS within +/- 50 mils
DQS1_P / DQS1_N	Byte1 DQS	CK_P - 40 ps	CK_P + 40 ps	[P&N] with +/- 5 mils	Within CK_P +/- 250 mils
DM1 DQ[15:8]	Byte1 Data	DQS1P - 8 ps	DQS1P + 8 ps		DQ / DM should be in reference to the corresponding DQS within +/- 50 mils
DQS2_P / DQS2_N	Byte2 DQS	CK_P - 40 ps	CK_P + 40 ps	[P&N] with +/- 5 mils	Within CK_P +/- 250 mils
DM2 DQ[23:16]	Byte2 Data	DQS2P - 8 ps	DQS2P + 8 ps		DQ / DM should be in reference to the corresponding DQS within +/- 50 mils
DQS3_P / DQS3_N	Byte3 DQS	CK_P - 40 ps	CK_P + 40 ps	[P&N] with +/-5 mils	Within CK_P +/- 250 mils
DM3 DQ[31:24]	Byte3 Data	DQS3P - 8 ps	DQS3P + 8 ps		DQ / DM should be in reference to the corresponding DQS within +/- 50 mils

Table 10-1. LPDDR4 Routing Recommendations.

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10.3.5 DRAM: Internal Length

Correct DRAM module layout requires consideration of the CV22FS chip internal lengths. Refer to the following table for the internal trace lengths of the 14 mm x 14 mm package.

10.3.5.1 Trace Length: 14 mm x 14 mm Package

Pin	Length (microns)
DDRE_DQS_BAR_0_LP4	6784.618
DDRE_DQS_0_LP4	7070.39
DDRE_DQS_BAR_1_LP4	4824.169
DDRE_DQS_1_LP4	4937.573
DDRE_CK_A_BAR_LP4	5868.328
DDRE_CK_A_LP4	6037.305
DDR0_DQS_BAR_0_LP4	7646.434
DDR0_DQS_0_LP4	7341.379
DDR0_DQS_BAR_1_LP4	7451.834
DDR0_DQS_1_LP4	7514.648
DDR0_CK_A_LP4	5284.644
DDR0_CK_A_BAR_LP4	5186.967
DDR0_CK_B_LP4	4308.457
DDR0_CK_B_BAR_LP4	4422.391
DDR0_DQS_BAR_2_LP4	7065.192
DDR0_DQS_2_LP4	6847.351
DDR0_DQS_BAR_3_LP4	7183.306
DDR0_DQS_3_LP4	7149.773
USB_DP	3657.362
USB_DM	3643.499
MIPI_DSI0_DP_0	6084.105
MIPI_DSI0_DN_0	6072.562
MIPI_DSI0_DP_1	5980.413
MIPI_DSI0_DN_1	5747.967
MIPI_DSI0_DP_2	3712.448
MIPI_DSI0_DN_2	3530.257
MIPI_DSI0_DP_3	6161.79
MIPI_DSI0_DN_3	6114.134
MIPI_DSI0_DP_CLK	3976.361
MIPI_DSI0_DN_CLK	3914.077
MIPI_DSI1_DP_0	4687.052
MIPI_DSI1_DN_0	4886.752
MIPI_DSI1_DP_1	1858.959
MIPI_DSI1_DN_1	1788.636
MIPI_DSI1_DP_2	2093.131
MIPI_DSI1_DN_2	2190.101
MIPI_DSI1_DP_3	3001.385
MIPI_DSI1_DN_3	2864.363
MIPI_DSI1_DP_CLK	2752.745
MIPI_DSI1_DN_CLK	2866.906

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Pin	Length (microns)
SD_LVDS_P_0	4880.764
SD_LVDS_N_0	5025.915
SD_LVDS_P_1	5973.898
SD_LVDS_N_1	5691.501
SD_LVDS_P_2	5113.338
SD_LVDS_N_2	5165.033
SD_LVDS_P_3	4126.686
SD_LVDS_N_3	3920.548
SD_LVDS_P_4	3949.131
SD_LVDS_N_4	3769.696
SD_LVDS_P_5	3599.339
SD_LVDS_N_5	3508.557
SD_LVDS_P_6	3886.014
SD_LVDS_N_6	3898.647
SD_LVDS_P_7	3305.347
SD_LVDS_N_7	3150.636
SD_LVDS_P_8	2777.592
SD_LVDS_N_8	2725.161
SD_LVDS_P_9	3363.974
SD_LVDS_N_9	3456.783
SD_LVDS_P_10	3885.055
SD_LVDS_N_10	3936.079
SD_LVDS_P_11	3942.54
SD_LVDS_N_11	4038.37
SPCLK_LVDS_P_0	4332.904
SPCLK_LVDS_N_0	4219.579
SPCLK_LVDS_P_1	3658.57
SPCLK_LVDS_N_1	3534.656
SPCLK_LVDS_P_2	2736.75
SPCLK_LVDS_N_2	2793.074
DDRE_CK_A_BAR_LP4	5868.328
DDRE_CK_A_LP4	6037.305
DDRE_CS_A_LP4	7795.539
DDRE_CS_2_A_LP4	8567.99
DDRE_CA_5_A_LP4	5565.091
DDRE_CA_4_A_LP4	5487.86
DDRE_CA_3_A_LP4	6437.014
DDRE_CA_2_A_LP4	7048.55
DDRE_CA_1_A_LP4	7469.55
DDRE_CA_0_A_LP4	8362.044
DDRE_DQS_BAR_0_LP4	6784.618
DDRE_DQS_0_LP4	7070.39
DDRE_DQ_7_LP4	5140.692
DDRE_DQ_6_LP4	6289.425
DDRE_DQ_5_LP4	5182.688
DDRE_DQ_4_LP4	4658.586
DDRE_DQ_3_LP4	4841.234
DDRE_DQ_2_LP4	5568.569

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Pin	Length (microns)
DDRE_DQ_1_LP4	6761.116
DDRE_DQ_0_LP4	6070.933
DDRE_DM_0_LP4	5869.2
DDRE_DQS_BAR_1_LP4	4824.169
DDRE_DQS_1_LP4	4937.573
DDRE_DQ_15_LP4	4682.057
DDRE_DQ_14_LP4	6488.287
DDRE_DQ_13_LP4	4419.814
DDRE_DQ_12_LP4	5100.315
DDRE_DQ_11_LP4	5546.997
DDRE_DQ_10_LP4	4531.094
DDRE_DQ_9_LP4	4405.041
DDRE_DQ_8_LP4	4443.296
DDRE_DM_1_LP4	5251.035
DDR0_DQ_7_LP4	8065.928
DDR0_DQ_6_LP4	7980.601
DDR0_DQ_5_LP4	5777.17
DDR0_DQ_4_LP4	6282.433
DDR0_DQ_3_LP4	5728.088
DDR0_DQ_2_LP4	5477.279
DDR0_DQ_1_LP4	5738.077
DDR0_DQ_0_LP4	5848.574
DDR0_DM_0_LP4	6033.357
DDR0_DQ_15_LP4	8851.754
DDR0_DQ_14_LP4	8867.751
DDR0_DQ_13_LP4	7228.222
DDR0_DQ_12_LP4	7140.759
DDR0_DQ_11_LP4	6568.158
DDR0_DQ_10_LP4	7018.111
DDR0_DQ_9_LP4	7684.792
DDR0_DQ_8_LP4	7840.743
DDR0_DM_1_LP4	8061.434
DDR0_DQ_23_LP4	5843.423
DDR0_DQ_22_LP4	7438.671
DDR0_DQ_21_LP4	5659.594
DDR0_DQ_20_LP4	7106.639
DDR0_DQ_19_LP4	5443.829
DDR0_DQ_18_LP4	5969.883
DDR0_DQ_17_LP4	4970.903
DDR0_DQ_16_LP4	6045.727
DDR0_DM_2_LP4	7187.787
DDR0_DQ_31_LP4	5913.097
DDR0_DQ_30_LP4	5254.112
DDR0_DQ_29_LP4	5272.557
DDR0_DQ_28_LP4	4643.521
DDR0_DQ_27_LP4	3960.796
DDR0_DQ_26_LP4	3991.086
DDR0_DQ_25_LP4	6355.977

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Pin	Length (microns)
DDR0_DQ_24_LP4	5072.442
DDR0_DM_3_LP4	6071.499
DDR0_CS_B_LP4	3836.023
DDR0_CS_2_B_LP4	4218.13
DDR0_CA_5_B_LP4	3548.865
DDR0_CA_4_B_LP4	4298.713
DDR0_CA_3_B_LP4	4878.602
DDR0_CA_2_B_LP4	4523.475
DDR0_CA_1_B_LP4	4583.218
DDR0_CA_0_B_LP4	3918.503
DDR0_CS_A_LP4	4166.211
DDR0_CS_2_A_LP4	5837.487
DDR0_CA_5_A_LP4	5090.374
DDR0_CA_4_A_LP4	5889.064
DDR0_CA_3_A_LP4	4522.74
DDR0_CA_2_A_LP4	5209.977
DDR0_CA_1_A_LP4	5966.676
DDR0_CA_0_A_LP4	4227.162
CEHU_SAFETY_ERR_L_0	3429.513
CEHU_SAFETY_ERR_L_1	2323.82
BG_REXT	3401.284
CAN0_RX	1776.383
CAN0_TX	1658.75
CAN1_RX	2480.929
CAN1_TX	1465.324
CLK_AU	5077.796
CLK_SI	4337.855
CLK_SI_1	4041.349
DDR0_CALIBR_LP4	4970.343
DDR0_CKE_2_LP4_LP3	3977.266
DDR0_CKE_LP4_LP3	3409.362
DDR0_RESET_LP4_LP3	4832.012
DDRE_CALIBR_LP4	8974.804
DDRE_CKE_LP4_LP3	6875.645
DDRE_RESET_LP4_LP3	7371.419
DETECT_VBUS	2273.42
ENET1_CLK_RX	3947.577
ENET1_GTX_CLK	3946.71
ENET1_MDC	2856.928
ENET1_MDIO	3109.828
ENET1_PTP_PPS	2346.041
ENET1_RXDV	3739.038
ENET1_RXD_0	2102.152
ENET1_RXD_1	2182.695
ENET1_RXD_2	1601.925
ENET1_RXD_3	3686.679
ENET1_TXD_0	3230.684
ENET1_TXD_1	3770.445

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Pin	Length (microns)
ENET1_TXD_2	6608.374
ENET1_TXD_3	2639.958
ENET1_TXEN	3865.825
ENET_2ND_REF_CLK	2944.286
ENET_CLK_RX	3638.297
ENET_EXT_OSC_CLK	4000.207
ENET_GTX_CLK	3893.606
ENET_MDC	12135.631
ENET_MDIO	10351.054
ENET_PTP_PPS	11897.77
ENET_RXDV	11223.895
ENET_RXD_0	11645.573
ENET_RXD_1	9050.717
ENET_RXD_2	10464.72
ENET_RXD_3	9542.914
ENET_TXD_0	8279.815
ENET_TXD_1	9550.903
ENET_TXD_2	7071.175
ENET_TXD_3	5100.422
ENET_TXEN	11248.084
GPIO24	3552.45
GPIO25	2574.517
I2S_CLK	5028.057
I2S_SI_0	1474.326
I2S_SO_0	3773.501
I2S_WS	3688.574
IDCSCLK	2903.371
IDCSDATA	3028.45
JTAG_CLK	10583.726
JTAG_RST_L	4237.846
JTAG_TDI	8796.705
JTAG_TDO	9819.199
JTAG_TMS	3242.801
POR_L	5201.99
PWM0	2503.501
PWM1	4545.859
PWM2	1597.619
SD_HS_SEL	1075.384
SENSOR_RST	3813.584
SHSYNC	2363.711
SMIO_0	2617.933
SMIO_1	3083.595
SMIO_2	4167.984
SMIO_3	3436.108
SMIO_4	3889.019
SMIO_5	2748.504
SMIO_6	3917.598

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Pin	Length (microns)
SMIO_7	2534.791
SMIO_8	2825.936
SMIO_9	2956.582
SMIO_18	2204.755
SMIO_19	3218.955
SMIO_20	3588.935
SMIO_21	3765.289
SMIO_22	2981.019
SMIO_23	2550.464
SMIO_24	4055.103
SMIO_25	3611.575
SMIO_26	3660.077
SMIO_27	4174.317
SMIO_28	3057.671
SMIO_29	4063.194
SMIO_30	4554.093
SMIO_31	3841.046
SMIO_32	3234.428
SMIO_33	4000.752
SMIO_34	2106.509
SMIO_35	3395.954
SSI0CLK	3848.503
SSI0EN0	4296.768
SSI0EN1	3831.071
SSI0MISO	3287.628
SSI0MOSI	3545.079
SSI2CLK	1141.787
SSI2EN0	2611.219
SSI2MISO	3558.946
SSI2MOSI	3009.059
SSI3CLK	1934.339
SSI3EN0	1974.892
SSI3MISO	2453.065
SSI3MOSI	2994.095
SVSYNC	1748.574
TEST_MODE	1448.499
TIMER0	2992.864
UART0_AHB_CTS_N	6084.416
UART0_AHB_RTS_N	700.629
UART0_AHB_RX	5899.348
UART0_AHB_TX	4620.536
UART1_AHB_CTS_N	4765.917
UART1_AHB_RTS_N	2304.202
UART1_AHB_RX	1851.912
UART1_AHB_TX	2610.036
UART_APB_RX	5492.174
UART_APB_TX	3642.461
USB_REXT	2191.771

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Pin	Length (microns)
XIN	3685.129
XI_RTC	3501.305
XOUT	3063.59
XO_RTC	2814.943

Table 10-2. Internal Trace Length (14 mm x 14 mm Package).

10.4 PCB Layout: Sensor / Video Input

10.4.1 VIN: MIPI CSI Mode

- Signal list: **SD_LVDS_P/N_[0:11], SPCLK_LVDS_P/N_[0:2]**
  - Trace lengths are 4500 mils or less.
  - Length matching is to:
    - +/- 5 mils between P and N.
    - +/- 50 mils among pairs.
  - Impedance is 100 ohms +/- 10% for differential pairs.
  - Trace space depends on the differential impedance requirement.
  - Trace space to adjacent signal traces is at least two times trace width (mils).
- Always keep P and N routed in parallel.
- Route P/N with fewer vias (3 vias maximum) and avoid sharp corners.
- Use arcs at corners.
- P/N differential signal pair should not cross routes with different power/ground planes.
- The P/N differential signal pairs must be routed together and should not be parallel with other signal traces in order to minimize crosstalk.
- The trace space to clock signal should be at least three times trace width or greater.

10.5 PCB Layout: Ethernet

- Transmit signal list: Ethernet pins **ENET_TXD_[0:3]** and **ENET_TXEN**
  - Keep all signal traces to the same length (or same trace delay and impedance).
  - Length matching is to +/- 100 mils.
- Receive signal list: Ethernet pins **ENET_RX_ER, ENET_CRS_DV**, and **ENET_RXD[0:3]**
  - Keep all signal traces to the same length (or same trace delay and impedance).
  - Length matching is to +/- 100 mils.

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10.6 PCB Layout: SMIO

- Signal list: Refer to the CV22FS chip datasheet.
- Signal traces should be approximately the same length (or have the same trace delay and impedance).
- Length matching is to +/- 100 mils.

10.7 PCB Layout: USB

- Signal list: **USB_DP**, **USB_DM**.
  - Note:
    - **GPIO1** (OVT0) / **GPIO2** (PWR0) are internally used for USB overcurrent detection and USB power enable. For details about the iomux function, refer to the *CV22FS Datasheet*.
  - Trace lengths are 2000 mils or less.
  - Length matching is to +/- 5 mils between **DP** and **DM**.
  - Differential trace impedance is 90 ohms.
  - Trace space depends on the differential impedance request.
  - Trace space to adjacent signal traces is at least two times trace width (mils).
- Always keep P/M routes in parallel (higher priority than keeping traces length-matched).
- Route P/M to minimize vias (less than two) and corners.
- Use arcs at corners.
- P/M trace should not route over split reference planes.
- The P/M signal pair must be routed together and not parallel with other signal traces in order to minimize crosstalk.
- Trace space to clock signals should be at least three times trace width or greater.
- Do not route USB traces under crystal, oscillators, clock synthesizers, magnetic devices or clock related circuitry.
- USB traces should be at least 90 mils from the edge of the plane (power and ground).

10.8 PCB Layout: Oscillator

- Signal list: **XIN**
- Trace lengths should be 500 mils or less.
- Trace width is 5 mils or greater.
- Trace space to adjacent signal traces is at least three times trace width or greater.

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10.9 PCB Layout: Crystal

- Signal list: **XIN**, **XOUT**, **XI_RTC**, **XO_RTC**
- Trace lengths should be 500 mils or less.
- Input / output trace length difference should be 5 mils or less.
- Trace space to adjacent signal traces is at least three times trace width or greater.

10.10 PCB Layout: Power Rails

- Add appropriate filter for all analog and PLL power rails.
- Place the power source as closely as possible to the power balls as shown on the reference board layout.
- Do not use thin traces for power delivery (i.e., ensure that the traces are wide and thick).
- **VDD** and **VDDA** must be kept separate in all hardware design.

10.11 PCB Layout: Miscellaneous

- All ground planes in different layers should connect with vias every 1 cm to 1.5 cm.
- All power trace widths should be 60 mils or greater.
- Place at least one 22-uF decoupling capacitor within every 6-inch^2 area.
- Recommended trace spacing
  - Left space: 3 mils
  - Trace width: 3 mils
  - Right space: 3 mils

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11. SoC ENVIRONMENTAL CONSIDERATIONS

11.1 SoC Environmental Considerations: Overview

This chapter includes environmental considerations for the SoC as follows:

- (Section 11.2) SoC Environmental Considerations: Temperature Control
  - (Section 11.2.1) Temperature Monitoring and Safety Enhancement

11.2 SoC Environmental Considerations: Temperature Control

To ensure that the temperature of the SoC is within the - 40º C to 125º C specification, the external auto-grade temperature sensor is recommended to monitor the temperature at the top of the SoC package or heat sink. If the temperature is beyond the specified range, then the temperature sensor must send an error message to the system management MCU to either throttle down system activity or shut down the system entirely.

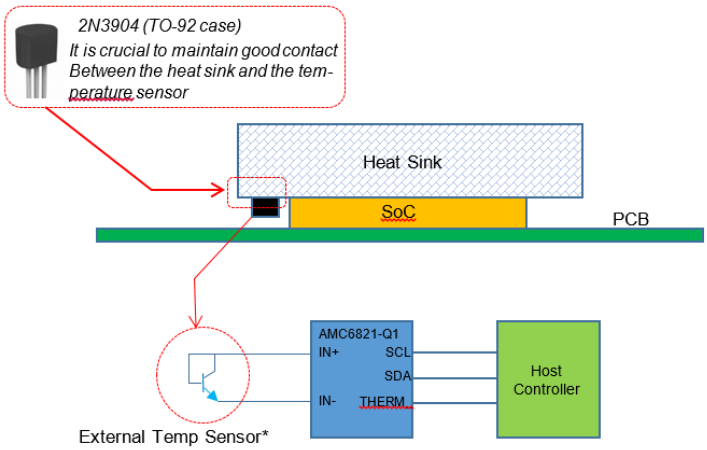
11.2.1 Temperature Monitoring and Safety Enhancement

In addition to the internal temperature sensor on the SoC, it is required to incorporate an external temperature sensor into the system board design, as it serves as a crucial backup for monitoring system temperatures and ensures that adequate cooling measures are in place. This redundancy is particularly beneficial in scenarios where the internal sensor fails or provides inaccurate readings due to system stress or other unforeseen issues. Most importanlty, this additional measure will enhance system safety and reliability.

For system design implementation, use an automotive-grade temperature monitor such as *AMC6821-Q1* or similar, along with a discrete *BJT 2N3904* as a remote temperature sensor.

To emulate the SoC's case temperature, perform the following:

1. Attach the remote sensor to the heat sink and position it close to the SoC, where it can accurately detect temperature changes (refer to the figure below).
2. Connect the sensor to the monitoring system.
3. Set up alerts for temperature thresholds to initiate appropriate action upon detecting a thermal event.



*For a detailed PCB layout of the external temperature sensor, consult the monitoring IC vendor

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NOTE: Page/chapter numbers for previous drafts may differ from those in the current version.

Version	Date	Comments
0.1	3 March 2020	Initial Version
0.2	29 May 2020	Updated Section 8.2 I2C / IDC Ports for port max Added Section 10.2 PCB Layout: System ESD Protection Mechanism Updated 10.3.4 DRAM: Signal Routing Example for LPDDR4 DRAM to add note Added SoC Environmental Considerations chapter
0.3	20 November 2020	Updated Table 6-4 DC Electrical Specifications Incorporated Rev A1 changes • Updated Figure 5-1 CV2FS Video Output Channels and Ports • Removed Section 5.3 VOUT: LVDS Interface
0.4	7 October 2021	Updated Figure 2-1 Power-on Sequence Timing for Power Rail Groups Added Section 2.4.3 Power-off Sequence Updated Section 3.2.8 POC: Boot Bypass Updated Section 3.2.3 POC: Boot Mode Summary for note Updated Chapter 4 Video Input to remove SLVS, LVDS, LVCMOS
0.5	12 December 2021	Updated Table 3-1 Power On Configuration Pin Assignments for bit 24 function Updated Section 2-3 RTC Disconnect Updated Table 3-4 Shared Boot Options Matrix to remove hs_boot Updated Section 2.4.2 Power-On Sequence Timing to add timing and circuit diagrams Updated Section 2.5 External Clock Source for Core and Real-Time Clock to remove a note Added Section 2.6 Watchdog Reset Updated Table 3-6 SPI NAND Boot Options for pin 20 description Updated Table 3-7 MMC Boot Options to remove hs_boot; updated pin 20 description Updated 3.2.7 POC: MMC Boot to add note Added Section 6.3 GPIO: Function Selection Added Section 6.3.1 Controlling the Boot Control Signals Added Section 6.4 GPIO: Alternations and Defaults Removed notes from previous Section 6.3 Updated Table 10-2 Internal Trace Length (14 mm x 14 mm) Package
0.6	25 August 2022	Added Section 6.9 GPIO: Note about the CAN Bus Updated Section 10.11 PCB Layout: Miscellaneous to add recommended trace spacing Removed Additional Resources chapter for consistency with other documentation
0.7	18 January 2023	Updated Table 3-7 MMC Boot Options for description of POC pin 17 Updated Section 6.5 GPIO: SD Interface Pull High / Low Requirements to add note
0.8	22 September 2023	Updated Section 6.2 GPIO: Characteristics to add non fail-safe note Added Section 6.10 GPIO: Note about SPI NOR
0.9	22 March 2024	Added Section 11.2.1 Temperature Monitoring and Safety Enhancement

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